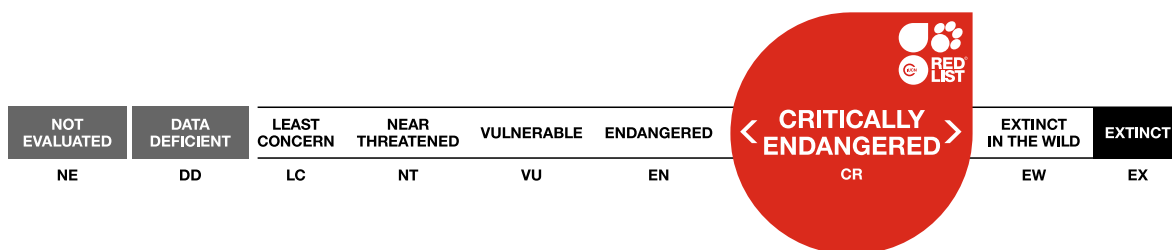


Desmana moschata, Russian Desman

Assessment by: Rutovskaya, M., Gazzard, A. & Turvey, S.T.



View on www.iucnredlist.org

Citation: Rutovskaya, M., Gazzard, A. & Turvey, S.T. 2023. *Desmana moschata*. The IUCN Red List of Threatened Species 2023: e.T6506A231334630. <https://dx.doi.org/10.2305/IUCN.UK.2023-1.RLTS.T6506A231334630.en>

Copyright: © 2023 International Union for Conservation of Nature and Natural Resources

Reproduction of this publication for educational or other non-commercial purposes is authorized without prior written permission from the copyright holder provided the source is fully acknowledged.

Reproduction of this publication for resale, reposting or other commercial purposes is prohibited without prior written permission from the copyright holder. For further details see [Terms of Use](#).

The IUCN Red List of Threatened Species™ is produced and managed by the [IUCN Global Species Programme](#), the [IUCN Species Survival Commission \(SSC\)](#) and [The IUCN Red List Partnership](#). The IUCN Red List Partners are: [ABQ BioPark](#); [Arizona State University](#); [BirdLife International](#); [Botanic Gardens Conservation International](#); [Conservation International](#); [Missouri Botanical Garden](#); [NatureServe](#); [Re:wild](#); [Royal Botanic Gardens, Kew](#); [Sapienza University of Rome](#); [Texas A&M University](#); and [Zoological Society of London](#).

If you see any errors or have any questions or suggestions on what is shown in this document, please provide us with [feedback](#) so that we can correct or extend the information provided.

Taxonomy

Kingdom	Phylum	Class	Order	Family
Animalia	Chordata	Mammalia	Eulipotyphla	Talpidae

Scientific Name: *Desmana moschata* (Linnaeus, 1758)

Regional Assessments:

- Europe

Common Name(s):

- English: Russian Desman
- Russian: vykhukhol'

Taxonomic Source(s):

ASM. 2021. Mammal Diversity Database (Version 1.9). Zenodo. <https://doi.org/10.5281/zenodo.4139722>. American Society of Mammalogists (ASM). Available at: www.mammaldiversity.org. (Accessed: 23 May 2022).

Taxonomic Notes:

This is the only representative of its genus.

Assessment Information

Red List Category & Criteria: Critically Endangered A2ac [ver 3.1](#)

Year Published: 2023

Date Assessed: March 29, 2023

Justification:

Surveys in Russia have indicated that the population has declined from at least 27,000 individuals in 2001 to 8,000–10,000 individuals in 2017. Based on these numbers, the decline by the year of assessment is estimated to have reached 83% over the previous 10 years. Whilst data are only available from the Russian extent of this species' range (which comprises the majority of its global range), it is likely that the data also reflect trends across the rest of its range in Ukraine and Kazakhstan, where populations are small and isolated. Threats include climate-driven changes in flood patterns, declining habitat quality, bycatch, water pollution and competition from introduced species, all of which remain present. Therefore, given the level of ongoing decline and threats faced by this species, it is assessed as Critically Endangered under criterion A2ac.

Previously Published Red List Assessments

[2016 – Endangered \(EN\)](#)

[2008 – Vulnerable \(VU\)](#)

1996 – Vulnerable (VU)

1994 – Vulnerable (V)

1990 – Vulnerable (V)

1988 – Vulnerable (V)

1986 – Vulnerable (V)

Geographic Range

Range Description:

The Russian Desman (*Desmana moschata*) occurs in Russia, Ukraine and Kazakhstan; previously, it was also present in Belarus, Lithuania (where it was introduced in the mid-1900s) (Balčiauskas 1996), and possibly the Republic of Moldova (see Genovesi *et al.* 2009), though its origins there are unclear. Its current range is very fragmented, and it has disappeared from many areas where it formerly occurred. At the beginning of the 20th century, the Russian Desman was common in the Dniepr, Don, Volga and Ural River basins. In Ukraine, historically it occurred in the basins of the Dnieper and the Don Rivers (Dniprovsk and Siversko-Donetska populations), but currently only sporadically occurs in the floodplain of the Seym River (Dnieper River basin) and its tributaries approximately 200 km west of the border with Russia (Mishta and Merzlikin 2009, Yemets *et al.* 2019). In Kazakhstan, it is thought to occur in Kostanay, the Ural River and the Kushum Channel, but recent data are lacking (Institute for Zoology and Gene Pool of Animals of National Academy of Scientists of Republic of Kazakhstan 1996).

Country Occurrence:

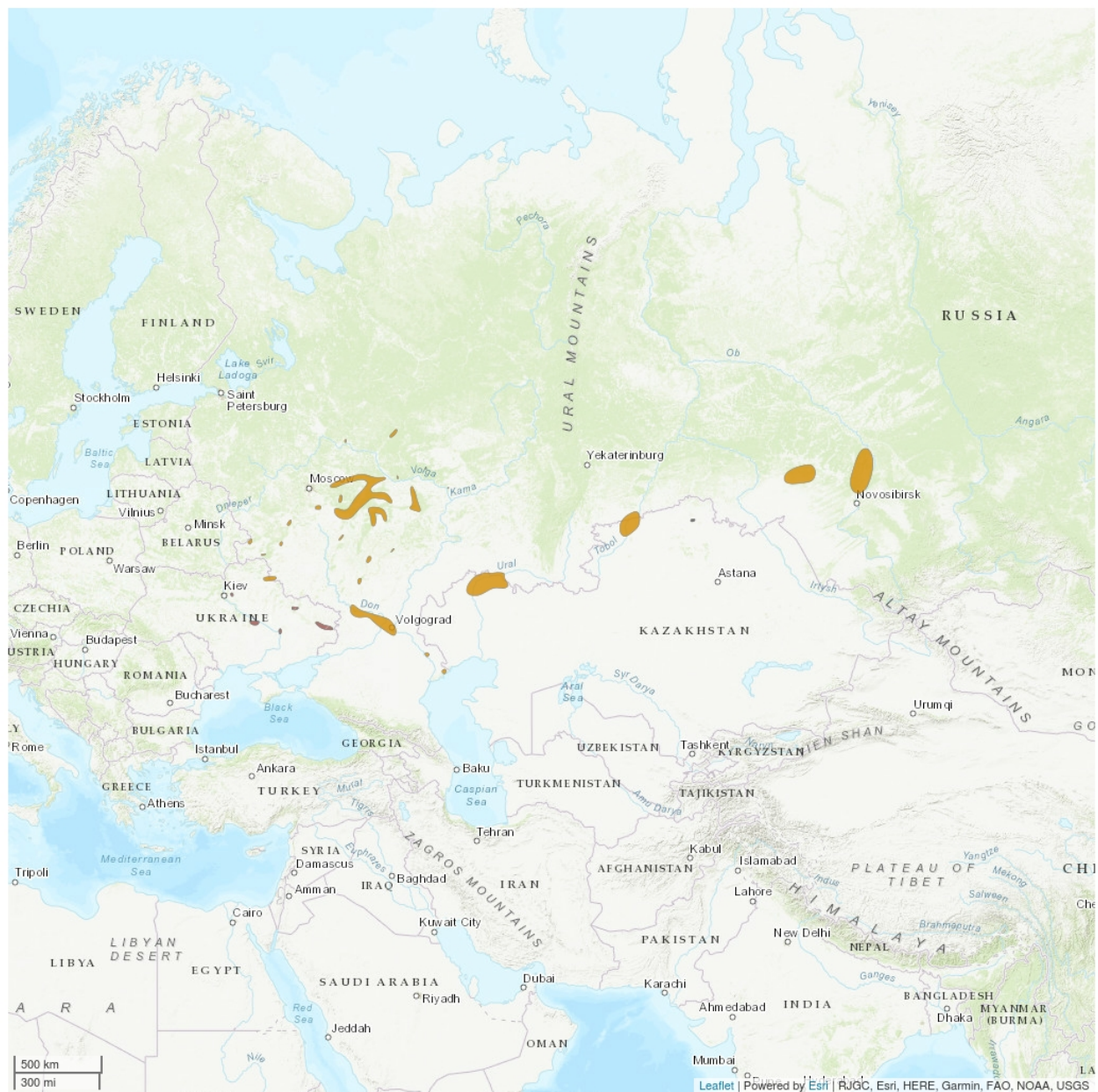
Native, Extant (resident): Kazakhstan; Russian Federation (European Russia); Ukraine

Native, Extinct: Belarus

Extinct & Introduced: Lithuania

Extinct & Origin Uncertain: Moldova

Distribution Map

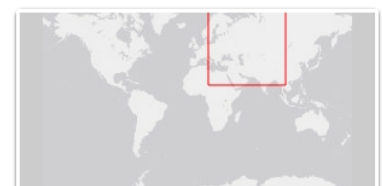
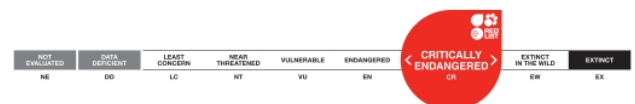


Legend

- EXTANT (RESIDENT)
- POSSIBLY EXTINCT
- PRESENCE UNCERTAIN

Compiled by:

IUCN SSC Small Mammal Specialist Group 2023



The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.

Population

At the beginning of the 1970s in the Soviet Union there were approximately 70,000 individuals, of which the majority occurred in Russia, with approximately 1,500 individuals in Kazakhstan and single records of occurrence in Ukraine and Belarus. Surveys in Russia indicate that the population there declined from 39,000 in 1985 to 27,000 by 2001 (see the Biodiversity Conservation Centre website <http://www.biodiversity.ru/eng/programs/desman/results.html> for further details). More recently, extensive burrow surveys undertaken during 2010–2016 have indicated a further decline: burrow density and the probable number of animals per burrow (which varies with month; Borodin 1963) were used to estimate relative abundance across 573 surveyed reservoirs, and then extrapolated to provide an estimate of 8,000–10,000 individuals remaining across Russia by 2017 (Rutovskaya *et al.* 2017). Assuming a linear ongoing decline, these estimates, when extrapolated, suggest that the population in 2023 will be around 3,600 individuals. This is consistent with recent population estimates of 4,000 or fewer (Rutovskaya 2022).

The Russian Desman has disappeared completely from some localities in Russia: surveys from 2009–2011 of apparently suitable habitat failed to detect any individuals in the Medveditsa River or in its tributaries and floodplain lakes, with the last reliable record for that area in 2008 within a tributary of the Medveditsa River (Oparina *et al.* 2013). However, a small population is thought to have survived in the Khopersky Reserve. In the Volga River basin, up to 3,500 individuals are concentrated in the Oka-Klyazma river system in Okskiy (M.V. Rutovskaya pers. comm. 2022). In the Ryazan region along the Oka River, the number of individuals is estimated to be 1,500. A population of up to 550 individuals lives in the lower reaches of the Sura River, another tributary of the Volga. In other areas where the desman has been reliably recorded, the number of local populations is estimated to comprise less than 100 individuals (Rutovskaya *et al.* 2017) and numbers are generally low. For example, the Klyzama River was formerly inhabited by one of the larger populations with 465 individuals recorded in 2012, but recent surveys have reported the presence of just 35 desmans (Zimenko and Rutovskaya 2022).

In Ukraine, currently the Russian Desman is very rare, potentially numbering a few hundred animals in one region (Mishta and Merzlikin 2009, Yemets *et al.* 2020). Its status in Kazakhstan is unknown. It is thought to now be locally extinct in Belarus, Lithuania and the Republic of Moldova.

Current Population Trend: Decreasing

Habitat and Ecology (see Appendix for additional information)

An aquatic species, it is found in oxbow lakes (with an area of 0.3 ha or more), and less frequently in ponds and rivers with slow flow, where it lives in holes in the banks. It does not inhabit all water bodies within its range; it has relatively strict habitat requirements and prefers water bodies 2–6 m in depth with rich water-marsh vegetation, bushes and primary forests along the banks, and with high macroinvertebrate abundance (Rutovskaya *et al.* 2020). In favourable years it is able to reproduce in any period of the year, but usually has two reproduction peaks at the end of spring and autumn with a brood of 1–6 young (more often 3–4) (Mishta and Merzlikin 2009). It is omnivorous, recorded feeding on water invertebrates and plant species (Kryštufek and Motokawa 2018).

Systems: Terrestrial, Freshwater (=Inland waters)

Use and Trade

This species now has no commercial or local use due to its low abundance. In the 19th and 20th centuries, it was caught extensively for fur and musk, which was used in perfumery. Desman trapping is now prohibited.

Threats (see Appendix for additional information)

The widespread use of stationary fishing nets—prohibited by law in many regions—is a serious threat to the Russian Desman. The very low price and high durability of modern nets means that poachers often leave them in the water for days or even months, checking on them only occasionally and often abandoning them. Russian Desmans will die, on average, within 5–10 minutes of being trapped in a net (Rutovskaya *et al.* 2017). Another key threat is the use of illegal 'electric landing nets' (or electric rods), which use an electric current to stun fish (Yemets *et al.* 2020); these items, also used by poachers, have become widespread over recent decades. They are not believed to directly harm desmans as a general rule, but can deplete populations of fish and aquatic invertebrates that the Russian Desman depends upon.

A further major threat to the Russian Desman is habitat loss and degradation. During the second half of the 20th century, water pollution, the creation of impoundments, drainage, clearance of riparian vegetation, and uncontrolled agricultural exploitation of flood plains became widespread and contributed to the decline in the species' population (M. Rutovskaya pers. comm. 2022). However, this process has abated somewhat in the last decade. Habitat and/or weather change as a result of climate change could be having a further marked impact on desman numbers, through modifying water flow regimes and water levels across suitable desman habitat. The absence of spring floods (which play an important role in the productivity of floodplain reservoirs, and in the success of desman breeding), and increasingly frequent and extremely destructive winter floods, as well as shallowing or drying up of floodplain reservoirs, is thought to have resulted in significant losses of individuals (Rutovskaya 2022).

The introduced American Mink (*Neovison vison*) may prey on the Russian Desman. Competition for breeding sites with Muskrats (*Ondatra zibethicus*) may also be a threat. The introduced Chinese Sleeper (*Perccottus glenii*) competes for similar food resources as the desman and can reduce the availability of vegetation and invertebrates (Podshivalina *et al.* 2020).

Conservation Actions (see Appendix for additional information)

The Russian Desman is listed on Appendix II of the Bern Convention and is included in the Red Data Book in Ukraine (declining), Russia (under category 1: population decreased to critical numbers) and Kazakhstan (category 2: vulnerable). In Russia, it is protected in the buffer zone of Okskiy, Khopersky, Prisursky, and Ugra National Parks and several small protected areas. This species has been the subject of multiple reintroduction attempts. For example, a reintroduction of 51 individuals was undertaken in 2001–2002 in a tributary of the Volga river basin, but in 2017 the population was estimated to consist of only 6–10 individuals (Bakka *et al.* 2018). In 2002–2003, it was also reintroduced to the Desna basin within Bryansky Les Reserve, Orlovskoe Polissya National Park, and Kerzhensky Reserve, but these reintroduced populations also remain small (Rutovskaya *et al.* 2017). Attempts to breed this species in captivity have been ongoing since the mid-20th century, but have failed (Rutovskaya *et al.* 2017).

Conservation measures recommended to stabilize and improve the condition of the Russian Desman

throughout its range include: the establishment of breeding populations in captivity; termination of imports of netting materials into Russia; strengthening and expanding inspections of regionally protected areas; and identification of appropriate habitat management and restoration regimes (Zimenko and Rutovskaya 2022).

Credits

Assessor(s): Rutovskaya, M., Gazzard, A. & Turvey, S.T.

Reviewer(s): Kennerley, R.

Contributor(s): Zuban, I., Koshkina, A., Mishta, A. & Kennerley, R.

Authority/Authorities: IUCN SSC Small Mammal Specialist Group

Bibliography

Andreychev, A., Kuznetsov, V. and Lapshin, A. 2019. Distribution and population density of the Russian desman (*Desmana moschata* L., Talpidae, Insectivora) in the Middle Volga of Russia. *Forestry Studies* 71: 48-68.

Bakka, S.V., Yu Kiseleva, N., Pankratov, I.I., Tarasov, I.A. and Shukov, P.M. 2018. The story of the creation and monitoring of the Russian Desman (*Desmana moschata* L.) population reintroduced of in the Kerzhenets river floodplain in the Nizhny Novgorod region. *IOP Conf. Series: Earth and Environmental Science* 115: 012036.

Balčiauskas, L. 1996. Lithuanian mammal fauna review. *Hystrix* 8: 9-15.

Borodin L.P. 1963. *Russian desman*. Book Publishing House, Saransk: Mordovia.

Genovesi, P., Bacher, S., Kobelt, M., Pascal, M. and Scalera, R. 2009. Alien Mammals of Europe. In: Delivering Alien Invasive Species Inventories for Europe (ed.), *Handbook of Alien Species in Europe*, pp. 399. Springer Nature.

Institute for Zoology and Gene Pool of Animals of National Academy of Sciences of Republic of Kazakhstan. 1996. The Red List of Kazakhstan. Available at: <http://redbook.kz/>.

IUCN. 2023. The IUCN Red List of Threatened Species. Version 2023-1. Available at: www.iucnredlist.org. (Accessed: 11 December 2023).

Kryštufek, B and Motokawa, M. 2018. Family Talpidae (moles, desmans, star-nosed moles and shrew moles). In: R.A. Mittermeier and D.E. Wilson (eds), *Handbook of the Mammals of the World. Insectivores, sloths and colugos*, Lynx Edicions, Barcelona.

Mishta, A. and Merzlikin, I. 2009. The Russian Desman *Desmana moschata* (Linnaeus 1758). In: Akimov, I.A. (ed.), *Red Book of Ukraine*, pp. 600. National Academy of Sciences, Kiev.

Oparina, O.S., Filinova, E.I., Sonina, E.E., Malinina Y.A., and Oparin, M.L. 2013. Current Status of the Russian Desman Habitats in Small Rivers of the Don River Basin in Saratov Oblast and the Abundance of This Species. *Biology Bulletin* 40(10): 854-861.

Pacifici, M., Santini, L., Di Marco, M., Baisero, D., Francucci, L., Grottolo Marasini, G., Visconti, P. and Rondinini, C. 2013. Generation length for mammals. *Nature Conservation* 5: 87–94.

Pavlov, D.S. (ed). 2021. *Red Data Book of the Russian Federation. Animals*. VNII Ecology.

Podshivalina, V.N., Osipov, V.V. and Egorov, L.V. 2020. The influence of benthivore species on foraging by the Russian desman *Desmana moschata* (Mammalia, Eulipotyphla) in water bodies of the lower Sura River (middle Volga region) (in Russian). *Bulletin of the Moscow Society of Naturalists* 125: 3-10.

Rutovskaya, M.V. 2022. On the change in the state of the population of the Russian desman over the past decade. Mammals in a changing world: current problems of theriology. XI Congress of the Theriology Society at the Russian Academy of Sciences. Moscow.

Rutovskaya, M.V., Aleksandrov, A.N., Podshivalina, V.N., Soboleva, A.S. and Glushenkov, O.V. 2020. Habitat conditions of *Desmana moschata* (Talpidae, Eulipotyphla, Mammalia) in the buffer zone of the Prisurskiy State Nature Reserve (Russia). *Nature Conservation Research* 5: 36-46.

Rutovskaya, M.V., Onufrenya, M.V. and Onufrenya, A.S. 2017. Russian desman (*Desmana moschata*: Talpidae) at the edge of disappearance. *Nature Conservation Research* 2: 100-112.

Yemets, O.M., Vlasenko, V.A., Demenko, V.M., Tatarynova, V.I., Rozhkova, T.O., Burdulaniuk, A.O., Bakumenko, O.M., Osmachko, O.M. and Shcherbyna, Y.M. 2019. Seymska population of Russian Desman (*Desmana moschata* L.) in north-eastern part of Ukraine: a history of formation and current state. *Indian Journal of Ecology* 47: 1077-1083.

Zimenko, A.V. and Rutovskaya, M.V. 2022. Urgent measures for the conservation of the Russian Desman. Mammals in a changing world: current problems of theriology. XI Congress of the Theriology Society at the Russian Academy of Sciences. Moscow.

Citation

Rutovskaya, M., Gazzard, A. & Turvey, S.T. 2023. *Desmana moschata*. *The IUCN Red List of Threatened Species* 2023: e.T6506A231334630. <https://dx.doi.org/10.2305/IUCN.UK.2023-1.RLTS.T6506A231334630.en>

Disclaimer

To make use of this information, please check the [Terms of Use](#).

External Resources

For [Supplementary Material](#), and for [Images and External Links to Additional Information](#), please see the Red List website.

Appendix

Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Habitat	Season	Suitability	Major Importance?
5. Wetlands (inland) -> 5.1. Wetlands (inland) - Permanent Rivers/Streams/Creeks (includes waterfalls)	Resident	Suitable	Yes
5. Wetlands (inland) -> 5.5. Wetlands (inland) - Permanent Freshwater Lakes (over 8ha)	Resident	Suitable	Yes
5. Wetlands (inland) -> 5.7. Wetlands (inland) - Permanent Freshwater Marshes/Pools (under 8ha)	Resident	Suitable	Yes

Use and Trade

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

End Use	Local	National	International
6. Other chemicals	No	No	No
10. Wearing apparel, accessories	No	No	No

Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Threat	Timing	Scope	Severity
2. Agriculture & aquaculture -> 2.1. Annual & perennial non-timber crops -> 2.1.4. Scale Unknown/Unrecorded	Ongoing	Unknown	Unknown
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation	
5. Biological resource use -> 5.4. Fishing & harvesting aquatic resources -> 5.4.3. Unintentional effects: (subsistence/small scale) [harvest]	Ongoing	Unknown	Unknown
	Stresses:	2. Species Stresses -> 2.1. Species mortality	
7. Natural system modifications -> 7.2. Dams & water management/use -> 7.2.8. Abstraction of ground water (unknown use)	Ongoing	Unknown	Unknown
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation	
7. Natural system modifications -> 7.3. Other ecosystem modifications	Ongoing	Unknown	Unknown
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation	
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (<i>Ondatra zibethicus</i>)	Ongoing	Unknown	Unknown

	Stresses:	2. Species Stresses -> 2.3. Indirect species effects -> 2.3.2. Competition -> 2.3.8. Other	
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (Neovison vison)	Ongoing	Unknown	Unknown
	Stresses:	2. Species Stresses -> 2.1. Species mortality	
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (Perccottus glenii)	Ongoing	Unknown	Unknown
	Stresses:	2. Species Stresses -> 2.3. Indirect species effects -> 2.3.2. Competition	
9. Pollution -> 9.3. Agricultural & forestry effluents -> 9.3.4. Type Unknown/Unrecorded	Ongoing	Unknown	Unknown
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation	
11. Climate change & severe weather -> 11.2. Droughts	Ongoing	Unknown	Unknown
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation	
11. Climate change & severe weather -> 11.4. Storms & flooding	Ongoing	Unknown	Unknown
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation	

Conservation Actions in Place

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Action in Place
In-place land/water protection
Conservation sites identified: Yes, over entire range
Occurs in at least one protected area: Yes
In-place species management
Successfully reintroduced or introduced benignly: Yes
In-place education
Subject to recent education and awareness programmes: Yes

Conservation Actions Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Action Needed	Notes
1. Land/water protection -> 1.1. Site/area protection	-
1. Land/water protection -> 1.2. Resource & habitat protection	-
2. Land/water management -> 2.1. Site/area management	-
2. Land/water management -> 2.2. Invasive/problematic species control	-
2. Land/water management -> 2.3. Habitat & natural process restoration	-

Conservation Action Needed	Notes
4. Education & awareness -> 4.3. Awareness & communications	-
5. Law & policy -> 5.4. Compliance and enforcement -> 5.4.4. Scale unspecified	-

Research Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Research Needed	Notes
1. Research -> 1.2. Population size, distribution & trends	-
1. Research -> 1.5. Threats	-
1. Research -> 1.6. Actions	-
2. Conservation Planning -> 2.1. Species Action/Recovery Plan	-
3. Monitoring -> 3.1. Population trends	-
3. Monitoring -> 3.4. Habitat trends	-

Additional Data Fields

Distribution
Estimated extent of occurrence (EOO) (km ²): 2820434
Continuing decline in extent of occurrence (EOO): Yes
Population
Continuing decline of mature individuals: Yes
Population severely fragmented: Yes
Habitats and Ecology
Continuing decline in area, extent and/or quality of habitat: Yes
Generation Length (years): 2

The IUCN Red List Partnership



The IUCN Red List of Threatened Species™ is produced and managed by the [IUCN Global Species Programme](#), the [IUCN Species Survival Commission \(SSC\)](#) and [The IUCN Red List Partnership](#).

The IUCN Red List Partners are: [ABQ BioPark](#); [Arizona State University](#); [BirdLife International](#); [Botanic Gardens Conservation International](#); [Conservation International](#); [Missouri Botanical Garden](#); [NatureServe](#); [Re:wild](#); [Royal Botanic Gardens, Kew](#); [Sapienza University of Rome](#); [Texas A&M University](#); and [Zoological Society of London](#).